

SPENCER MICHAELSON

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Portfolio: <https://smichaelson.github.io/>

EDUCATION

Harvey Mudd College, Claremont, CA
B.S. in Engineering | GPA: 3.912/4.000

May 2028

PROFESSIONAL EXPERIENCE

Engineering Intern III, Lean Technology Corporation, Costa Mesa, CA

May 2026 - August 2026

- *Promoted from Engineering Intern I (May 2025 - August 2025) and Engineering Intern II (August 2025 - May 2026).*
- Modeled **electrical fault scenarios** and **circuit-network behavior** for major airport infrastructure programs at SFO and SEA-TAC.
- Built a QGIS workflow integrating 360° field video, survey data, and geospatial layers to accelerate field verification, spatial analysis, and design coordination for airport electrification projects.
- Created Python/SQL project-management and electrical forecasting tools for a **\$1.6B SFO electrification and expansion portfolio**, supporting planning across 250+ infrastructure projects under V&V-driven workflows.
- Automated airport traffic modeling, regulatory/weather/RF/geologic data analysis, and FAA documentation workflows using Python and SQL, **reducing turnaround time by 5x**.

TECHNICAL LEADERSHIP

Chief Engineer, Mudd Amateur Rocketry Club (MARC), Harvey Mudd College

August 2024 - Present

- *Promoted from Structures Member (August 2024 - August 2025) and Propulsion Lead (August 2025 - March 2026).*
- Served as Chief Engineer for **Apollyon I**, a 6-inch, 60-lb, full-M-class **FAR Unlimited 2026** vehicle; led a 40-member team across propulsion, structures, avionics, simulation, recovery, payloads, integration, and launch operations, earning **1st place**.
- Owned vehicle architecture, requirements flowdown, integration schedule, technical reviews, and launch-readiness planning across propulsion, structures, avionics, recovery, payload, and flight simulation.
- Directed trade studies for motor selection, stability margin, max-Q, recovery sizing, and composite airframe design leveraging OpenRocket, RASAero II, MATLAB, SolidWorks, and COMSOL; downselected an architecture based on stability, recovery mass, and competition constraints.
- Fabricated and structurally integrated **Gladius III**, contributing to **3rd place at FAR Unlimited 2025**; validated flight simulation against altimeter data with **99.95% apogee prediction accuracy**.

Chief Technical Analyst, Claremont Aerospace & Defense Association (CADA), Harvey Mudd College

December 2025 - Present

- Led aerospace/defense technical analyses, translating stakeholder objectives into payload concepts, mission architectures, CONOPS, integration constraints, and risk assessments for autonomous system applications.
- Directed analysis of current arms manufacturing processes and developed an additional product stack while accounting for regulations, existing facilities, and competition.

President, Mudd Automotive Club Hybrids (MACH), Harvey Mudd College

August 2024 - Present

- *Promoted from Mechanical Member (August 2024 - August 2025) and Safety Officer (August 2025 - April 2026).*
- Lead a 13-member **hybrid-electric vehicle team** across mechanical design, electrical integration, manufacturing, safety, and testing workstreams.
- Oversee design reviews, the build schedule, and subsystem integration for chassis, braking, steering, and hybrid-powertrain systems.
- Designed and manufactured frame, braking, and steering assemblies in SolidWorks using DFM principles and FEA to evaluate safety factor, stiffness, and load paths.

SELECTED PROJECTS

Independent Project, **Hat Trick**, Harvey Mudd College

February 2025 - March 2026

- Designed, built, and flew **Hat Trick** through **6** high-power launch campaigns to 7,000 ft on K-class motors, validating recovery, avionics, and structural integration across **iterative flight tests**.
- Developed a **dual-redundant avionics stack** for flight data logging, deployment control, and recovery reliability using multi-sensor fusion, LoRa telemetry, and custom enclosures.
- Implemented onboard **GPS tracking and a post-flight data pipeline** to reconstruct trajectory, evaluate recovery performance, and compare flight data against simulation predictions.
- Analyzed recovered flight data and post-flight inspection results to improve avionics packaging, recovery reliability, and launch integration procedures.

Independent Project, **Unknown Rider**, Harvey Mudd College

March 2026 - Present

- Developing **Unknown Rider**, a NAR Level 3 **certification vehicle and avionics test platform** for high-altitude flight-data collection, recovery validation, and embedded-system testing.
- Performing preliminary aerodynamic, structural, and RF link-budget analyses to assess stability margin, airframe survivability, telemetry range, and recovery architecture for projected **50,000+ ft / Mach 2+** flight conditions.

Independent Project, **Atlas Flight Guidance Computer**, Harvey Mudd College

March 2026 - Present

- Designing **Atlas**, a custom STM32-based flight-guidance computer for state estimation, active control, data logging, recovery deployment, live telemetry, GPS tracking, peripheral expansion, and guidance-interface research.
- Manufactured the computer and wrote system firmware and an RTOS, confirming **full functionality** of actuator control, sensors, Bluetooth, LoRa, GNSS, USB-C, buzzer, local storage, and GPIO under load.

RESEARCH EXPERIENCE

Undergraduate Researcher, Harvey Mudd College: Drone Lab, Dr. Jason Gallichio

January 2025 - December 2025

- Created **drone guidance and flight-analysis tools** for radio telescope beam mapping, supporting autonomous flight planning, beam-prediction workflows, and post-flight trajectory analysis.
- Built a hardware-agnostic forward simulation to predict beam-mapping coverage and flight-path performance before field testing.
- Planned, executed, and analyzed **40+ autonomous flights**, tuning GNC behavior and evaluating trajectory repeatability using flight logs and post-processing scripts.

SKILLS

Programming: MATLAB (10 years), Python (7 years), R (6 years), C/C++ (5 years), SQL (3 years) | **Development tools:** Git (5 years)

Software: SolidWorks, Simulink, COMSOL, Bluebeam, AutoCAD, pgAdmin, QGIS, KiCad, STM32 Ecosystem, OpenRocket, RASAero II

Analysis: Electrical fault simulation, FEA, CFD, flight simulation, post-flight analysis, GNC, trajectory reconstruction

Manufacturing: Additive manufacturing, manual machining, CNC machining, laser cutting, water jet, composites, GD&T

Certifications: NAR Level 1 & NAR Level 2 High-Power Rocketry